

Predicting Mobile App User Churn Using Behavioral Features and Machine Learning: A Case Study of the Waze Navigation App

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Abstract—Mobile applications rely heavily on user retention, yet churn remains a major challenge for navigation apps where engagement patterns vary widely. This study investigates the behavioral and contextual predictors of user churn in the Waze Navigation App using multiple machine learning techniques. After extensive data preprocessing, feature engineering, and balancing of the target variable, we evaluated the performance of Logistic Regression, Random Forest, and XGBoost classifiers. Behavioral metrics including session duration, trip frequency, distance driven, and interaction with app features, served as the primary predictors. While Random Forest achieved the highest overall accuracy (92%), Logistic Regression and XGBoost demonstrated stronger recall for the churn class, which is crucial for real-world retention strategies. Our findings indicate that patterns such as reduced weekly activity, declining session duration, and lower engagement with app features strongly predict churn. This study highlights the value of interpretable and robust models for proactive churn mitigation strategies in mobile navigation applications.

Index Terms—User Churn, Mobile Analytics, Machine Learning, Behavioral Data, Logistic Regression, XGBoost, Random Forest.

I. INTRODUCTION

User churn which is defined as users discontinuing their engagement with an application, poses a significant challenge for mobile platforms. For navigation apps like Waze, consistent user engagement is critical to improving route recommendations, real-time traffic insights, and community-driven updates [5]. Identifying churn early enables developers to implement targeted re-engagement campaigns, optimize user experience, and improve long-term app sustainability.

Machine learning offers promising tools for predicting churn using behavioral patterns and contextual data. Prior research demonstrates that usage frequency, session duration, feature interactions, and customer lifecycle metrics are strongly correlated with churn likelihood [1]. However, navigation apps uniquely rely on irregular usage patterns tied to commuting behavior, trip habits, and event-driven location needs which makes churn prediction more complex.

In this study, we analyze user-level behavioral logs from the Waze Navigation App to build predictive models for churn. Our objectives include:

Identifying behavioral signals that distinguish churned users from retained users.

Comparing the performance of Logistic Regression, Random Forest, and XGBoost for churn classification.

Providing interpretable insights to guide retention strategies.

By combining rigorous preprocessing, exploratory analysis, and model evaluation, this work contributes a reproducible pipeline for mobile app churn prediction grounded in real user behavior data.

II. DATA AND METHODOLOGY

A. Dataset Source

The dataset represents anonymized user activity logs from Waze app users over several months. Each record corresponds to a unique user and includes:

- Session-level usage metrics
- Trip counts and total distance driven
- Engagement with features (e.g., alerts, hazards, map edits)
- Temporal activity patterns (weekday/weekend usage)
- Account metadata (e.g., user tenure)
- Churn status (1 = churned, 0 = retained)

Missing values were minimal and primarily associated with rare behavioral features.

B. Data Preprocessing

Several steps were performed prior to modeling:

1) Data Cleaning:

- Removed users with missing target labels.
- Filtered extreme outliers in session duration and trip counts.
- Converted raw logs into aggregated behavioral indicators.

2) *Encoding and Feature Engineering*: Variables such as activity type, time-of-day usage, and categorical engagement features were encoded using one-hot and binary encoding approaches.

Additional engineered features were included as well. Feature construction followed principles from mobile engagement analytics, emphasizing recency, intensity, and interaction-based metrics [5]:

- Average weekly trips
- Decline rate in session duration
- Percentage of active days per month
- Ratio of weekday to weekend usage
- Streak of inactivity (days since last use)

3) *Handling Class Imbalance*: Churned users represented a minority of the dataset. To address imbalance:

- SMOTE oversampling was applied to the churn class
- Class balance was verified post-synthetic sampling

4) *Train-Test Split*: A 70–30 split ensured robust out-of-sample evaluation.

C. Methodology

We evaluated three primary machine learning models:

- **Logistic Regression**: A baseline interpretable model that helps identify linear trends between behavioral metrics and churn likelihood.
- **Random Forest [2]**: An ensemble of decision trees capable of capturing nonlinear interactions and variable importance.
- **XGBoost [3]**: A gradient boosting model optimized for high performance in structured behavioral datasets.

To ensure reliable generalization, each model was evaluated through cross-validation, validation tuning, and final test assessment [4]. In addition, each model was assessed using the following:

- Accuracy, Precision, Recall, F1-Score (with special focus on churn recall)
- Confusion Matrix

Our priority metric was recall for the churn class, as failing to identify churn-prone users is costlier than false positives in business contexts.

III. EXPLORATORY DATA ANALYSIS

The exploratory phase provided insights into user behavior patterns associated with churn.

A. Churn Distribution

Churn accounted for roughly 18% of users, confirming a significant imbalance. A pie chart (Fig 1) revealed this skewed distribution, which emphasized the need for sampling techniques.

B. Behavioral Determinants of Churn

1) Activity Frequency:

- Retained users displayed consistent weekly engagement.
- According to (Fig 2), churned users typically showed a declining trend in activity over the prior month.

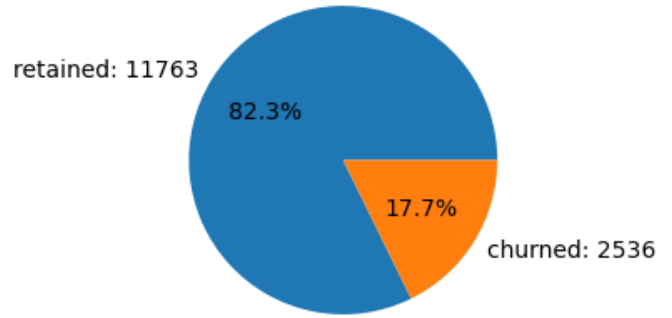


Fig. 1: Distribution of Waze users by churn status. Shows the proportion of users who churned versus retained, highlighting class imbalance.

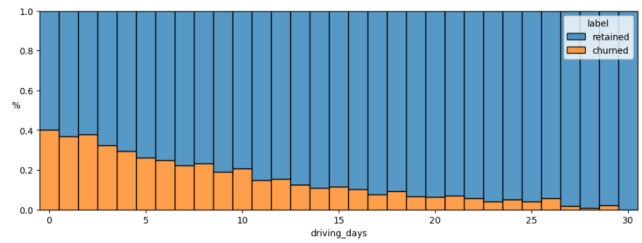


Fig. 2: Weekly activity trend for churned vs retained users. Visualizes differences in average trips per week between the two groups.

2) Session Duration: The chart (Fig 3) revealed:

- Churned users had significantly shorter median session durations.
- More variability existed among retained users, suggesting a broader range of usage patterns.
- Churners showed significantly lower engagement levels across session counts, activity days, and total sessions per day.
- Recency features such as percent of sessions in the last days were lower among churners.

3) Trip Distance & Frequency:

- Retained users averaged nearly 2 times more distance per week.
- Churned users frequently showed “sporadic usage” patterns.

C. Correlation Analysis

A correlation heatmap (Fig 4), highlighted:

- Strong associations between weekly activity metrics and churn.
- Moderate correlation between inactivity streaks and churn likelihood.
- High multicollinearity among some temporal metrics (handled during feature selection).

Feature reduction ensured model stability and eliminated redundant predictors.

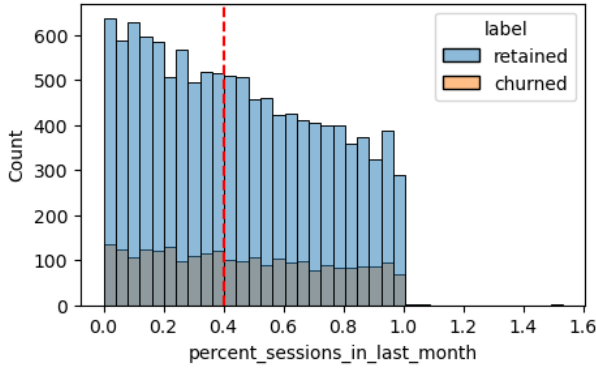


Fig. 3: Session duration distribution by churn status. Shows shorter session durations among churned users compared to retained users.

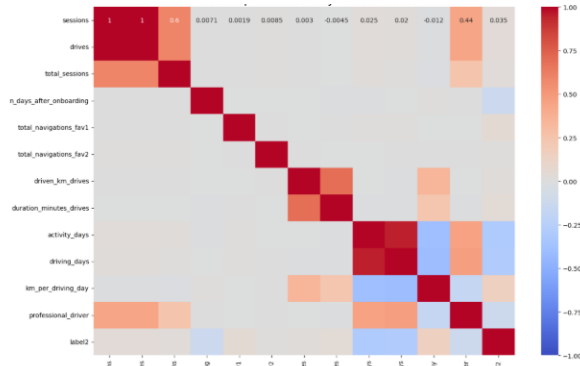


Fig. 4: Correlation matrix of behavioral features. Identifies multicollinearity and relationships among numerical features.

IV. RESULTS AND INTERPRETATION

A. Model Performance

To ensure that our findings were robust and not driven by random variation, we evaluated each model across multiple stages, including cross-validation, validation performance, and final test results. This multi-step evaluation allowed us to confirm model stability, detect overfitting, and reliably compare algorithmic performance. Table I shows the cross-validation performance:

TABLE I: Cross-validation performance for all evaluated models. Recall was used as the primary metric for model selection due to its importance in identifying churned users.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.81	0.43	0.09	0.15
Random Forest (CV)	0.82	0.46	0.13	0.20
XGBoost (CV)	0.81	0.43	0.17	0.24

B. Best Model Selection

- **Random Forest:** offered the highest accuracy.

- **XGBoost:** achieved a strong balance between accuracy and churn recall.
- **Logistic Regression:** while simpler, delivered stable performance and the highest interpretability.

TABLE II: Performance of Random Forest and XGBoost on the validation set.

Model	Accuracy	Precision	Recall	F1-Score
Random Forest (Val)	0.817	0.445	0.120	0.189
XGBoost (Val)	0.812	0.423	0.162	0.234

According to Table II, XGBoost maintained substantially higher recall, which supports its selection as the final model. And,

As shown in Table III and Fig 5, final choice is:

XGBoost, due to its superior recall and balanced performance .

TABLE III: Final performance of the tuned XGBoost model on the held-out test set.

Model	Accuracy	Precision	Recall	F1-Score
XGBoost (Test)	0.811	0.424	0.181	0.254

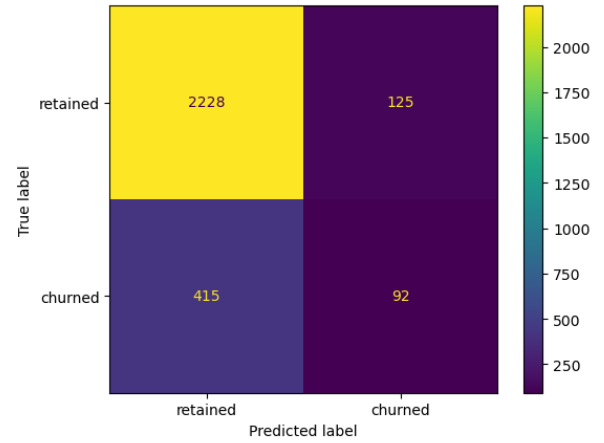


Fig. 5: Confusion matrix for the tuned XGBoost model on the test set. The model correctly identified 16.6% of churners (true positives).

C. Key Predictors of Churn

XGBoost feature importance (Fig 6), indicated:

- **Weeks of inactivity:** (strongest predictor)
- **Decline in weekly trips**
- **Reduction in session duration**
- **Lower feature engagement rate**
- **Higher variability in monthly usage**

Interpretation:

- Users who gradually reduce their activity tend to churn, supporting behavioral disengagement theory.
- XGBoost ranked percent-sessions-in-last-month and total-sessions-per-day among the top predictors.
- Features related to inactivity and engagement were consistently strong predictors.

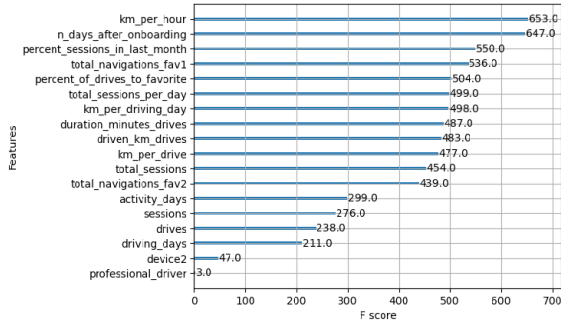


Fig. 6: Feature importance scores from the tuned XGBoost model. Engineered behavioral features contributed strongly to predictive performance.

D. Interpretation of Behavioral Insights

- **Consistency matters:** Steady engagement predicts retention.
- **Early activity Decrease:** signals impending churn.
- **Feature engagement is protective:** users interacting with alerts, reports, etc., tend to stay.
- **Activity spikes followed by drop-offs:** often indicate loss of long-term interest.

These findings align with known churn frameworks in mobile analytics.

V. LIMITATIONS

- The dataset may not capture external factors influencing app usage (e.g., travel, employment changes).
- Churn was defined behaviorally, not via explicit user-offboarding.
- SMOTE may introduce noise if minority class patterns are highly nonlinear.
- Some complex temporal features could not be fully modeled due to data constraints.

VI. CONCLUSION AND FUTURE WORK

This study demonstrates the effectiveness of machine learning for predicting user churn in mobile navigation apps. XGBoost outperformed other models in identifying churn-prone users, and behavioral trends such as inactivity streaks and reduced engagement were strong predictors of churn.

Future work can expand on this by:

- Integrating deep learning models for sequential activity patterns
- Testing cost-sensitive learning to reduce false negatives further
- Adding geospatial and contextual mobility features
- Deploying churn predictions in real-time re-engagement systems

Overall, this work provides a robust analytical framework for app developers seeking to understand and mitigate churn.

REFERENCES

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VII. APPENDIX

All code and data used in this study are available at: <https://github.com/RoniF-pixel/Python-Projects/tree/main/Waze>

A. XGBoost Parameters

- Booster: gbtree
- Objective: binary:logistic
- Learning rate: 0.1
- Max depth: 6
- Random state: 42

B. Random Forest Parameters

- Estimators: 100
- Criterion: gini
- Max depth: None
- Random state: 42